

COMMENTI SULLA FINE DELL'ESERCITAZIONE
PRECEDENTE

30/09/20

$$\vec{E}_{\text{disc}} = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right) \hat{x}$$

• $x \rightarrow 0$, $\vec{E}_{\text{disc}} \rightarrow \frac{\sigma}{2\epsilon_0} \hat{x}$ ($x \ll R$)

• $x \rightarrow \infty$ ($x \gg R$)

$$1 - \frac{x}{\sqrt{x^2 + R^2}} = 1 - \frac{x}{x \sqrt{1 + \left(\frac{R}{x}\right)^2}} , \quad y = \left(\frac{R}{x}\right)^2$$

$$(1+y)^{-1/2} \sim 1 - \frac{R^2}{2x^2} \quad \begin{matrix} (1+y)^\alpha \sim 1 + \alpha y \\ y \rightarrow 0 \quad [\alpha = -1/2] \end{matrix}$$

$$\vec{E}_{\text{disc}} \rightarrow \frac{\sigma}{2\epsilon_0} \cdot \frac{R^2}{2x^2} = \frac{Q}{2\epsilon_0 \pi R^2} \cdot \frac{R^2}{2x^2} = \frac{Q}{4\pi\epsilon_0} \cdot \frac{1}{x^2}$$

① Esercizio 1



$$\vec{E} = E(r) \cdot \vec{u}_r$$

$$\phi(\vec{E}) = \oint E(r) \vec{u}_r \cdot \vec{u}_n dS$$

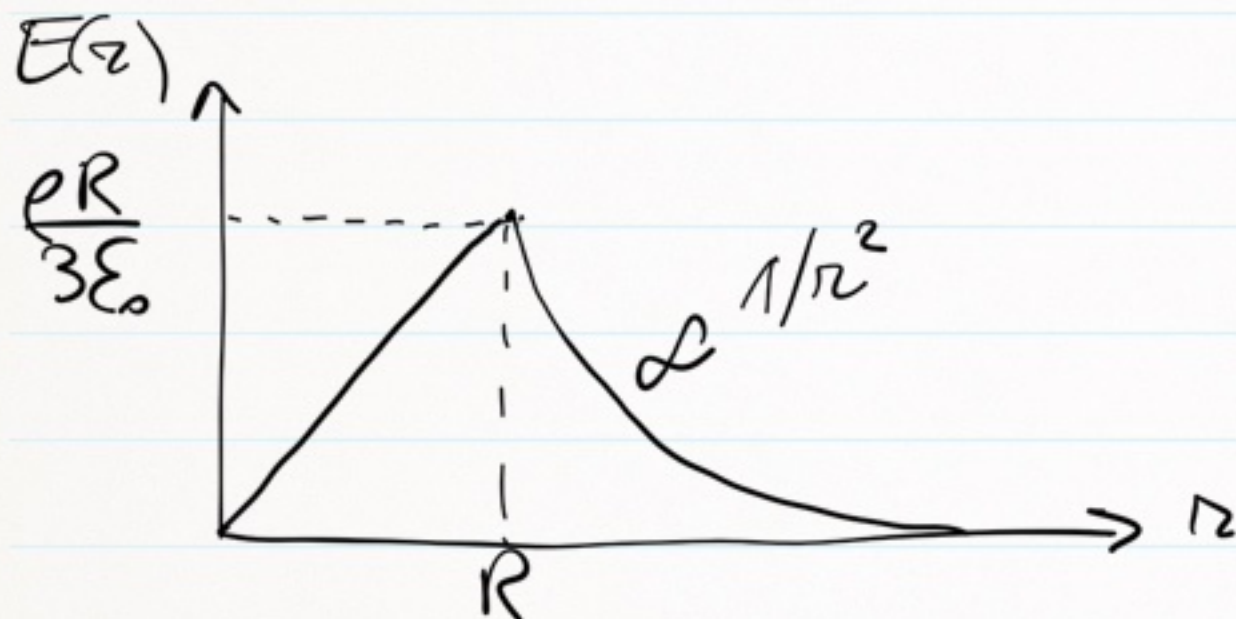
$$Q = ?$$

$$= E(r) \oint dS = E \cdot 4\pi r^2$$
$$\frac{1}{\epsilon_0} \frac{Q}{4\pi r^2}$$

$$r < R: Q = \rho \cdot \frac{4}{3}\pi r^3 \Rightarrow E(r) = \frac{\rho}{3\epsilon_0} \cdot r$$

$$r > R: Q = \rho \cdot \frac{4}{3}\pi R^3 \Rightarrow E(r) = \frac{\rho}{3\epsilon_0} \cdot \frac{R^3}{r^2} =$$

$$= \frac{Q}{4\pi\epsilon_0} \cdot \frac{1}{r^2} (!)$$



POTENZIALE ELETTROSTATICO (V)

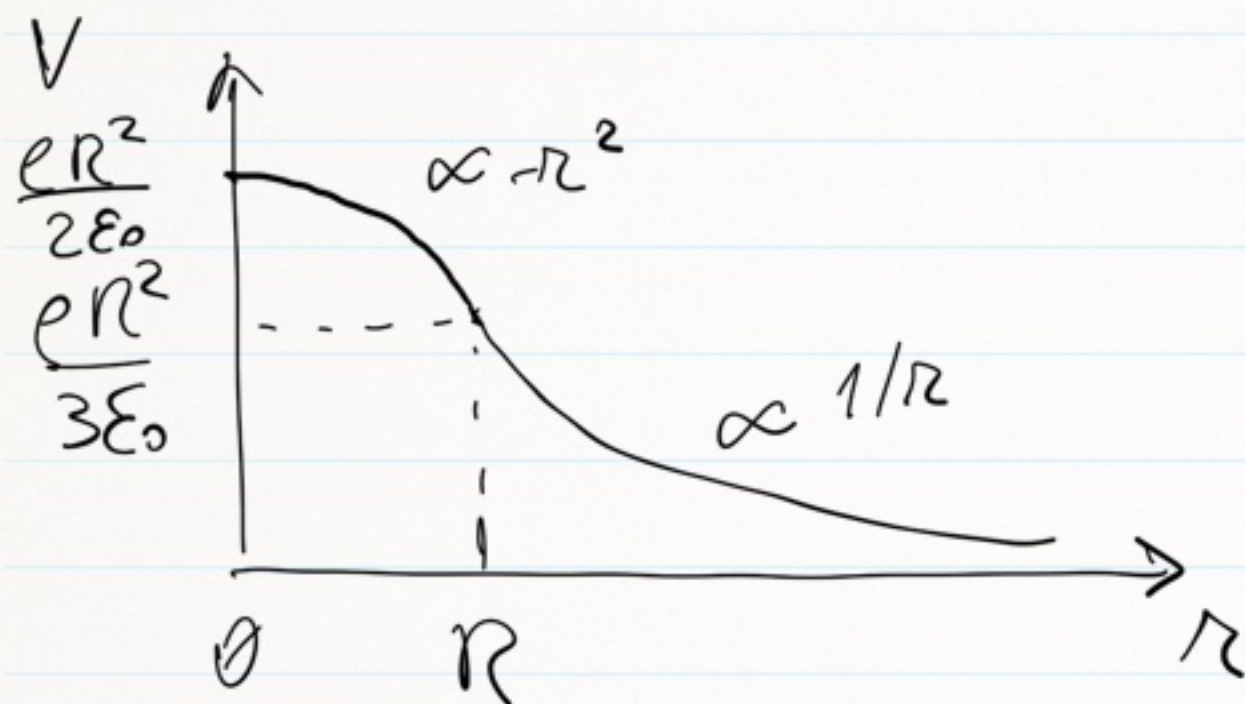
$$A \cdot \quad \cdot \quad \int_A^B [V_A - V_B = \int_A^B E(r) dr]$$

$$V \propto 1/r, \quad r \rightarrow \infty, \quad V(r) \rightarrow 0 \quad (!)$$

$$\begin{aligned} \text{• } \underbrace{r > R} \quad V(r) - \underbrace{V_\infty}_{\rightarrow 0} &= \int_r^\infty E(r') dr' = \int_r^\infty \frac{e}{3\epsilon_0} \frac{R^3}{r'^2} dr' = \\ &= \frac{e R^3}{3\epsilon_0} \left[-\frac{1}{r'} \right]_r^\infty = \frac{e R^3}{3\epsilon_0} \cdot \frac{1}{r} \end{aligned}$$

$$\text{• } \underline{r < R} : \quad V(r) - V(r=R) = \int_r^R E(r') dr'$$

$$\begin{aligned} V(r) &= \frac{e R^2}{3\epsilon_0} + \frac{e}{3\epsilon_0} \left[\frac{r'^2}{2} \right]_r^R = \frac{e}{3\epsilon_0} \left(R^2 + \frac{R^2 - r^2}{2} \right) = \\ &= \frac{e}{3\epsilon_0} \cdot \frac{3R^2 - r^2}{2} \end{aligned}$$



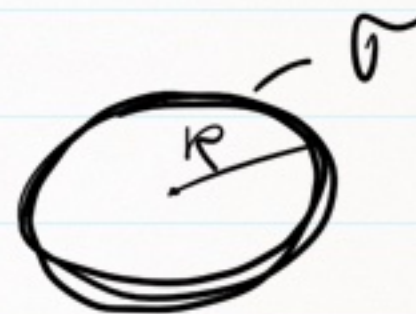
② Esercizio 2

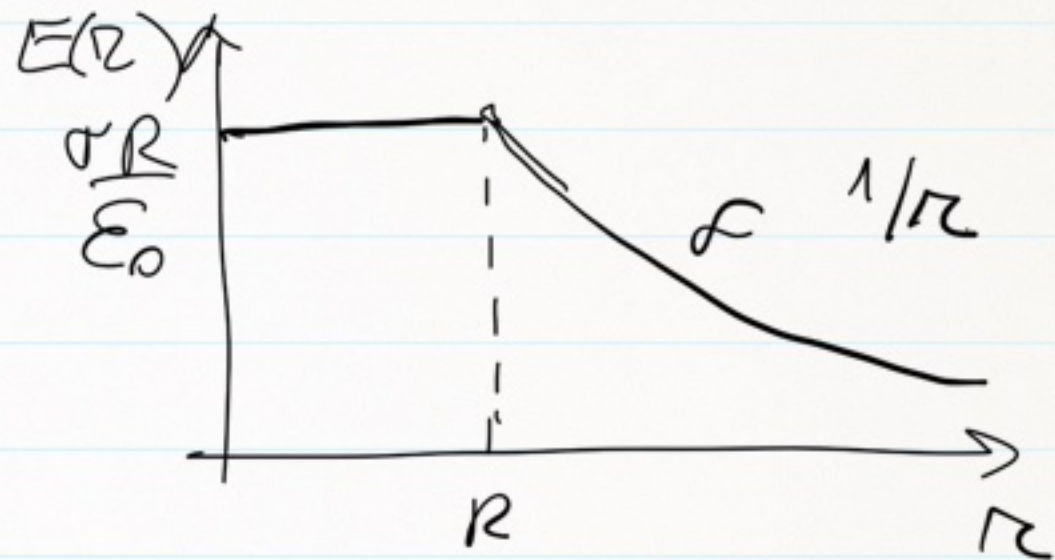
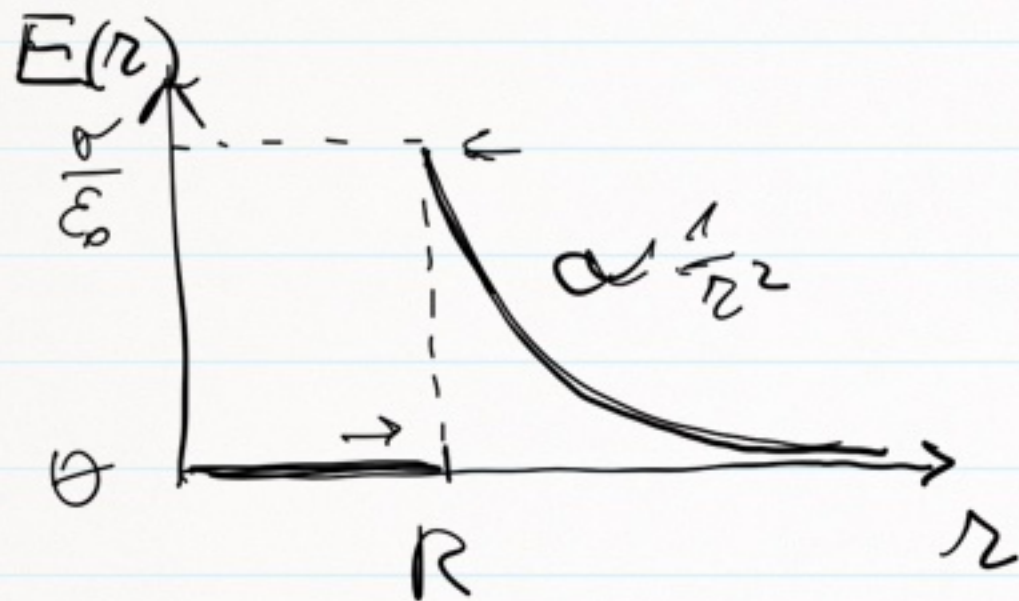
• CAMPO ELETTROSTATICO

$$r < R : E(r) = 0$$

$$r > R : E(r) \cdot 4\pi r^2 = \frac{\sigma \cdot 4\pi R^2}{\epsilon_0}$$

$$\Rightarrow E(r) = \frac{\sigma R^2}{\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon_0} \cdot \frac{1}{r^2} (!)$$





• POTENZIALE ELETTROSTATICO

$$r > R: \quad V(r) = \int_r^{\infty} E(r') dr' = \frac{\sigma R^2}{\epsilon_0} \left[-\frac{1}{r'} \right]_r^{\infty} =$$

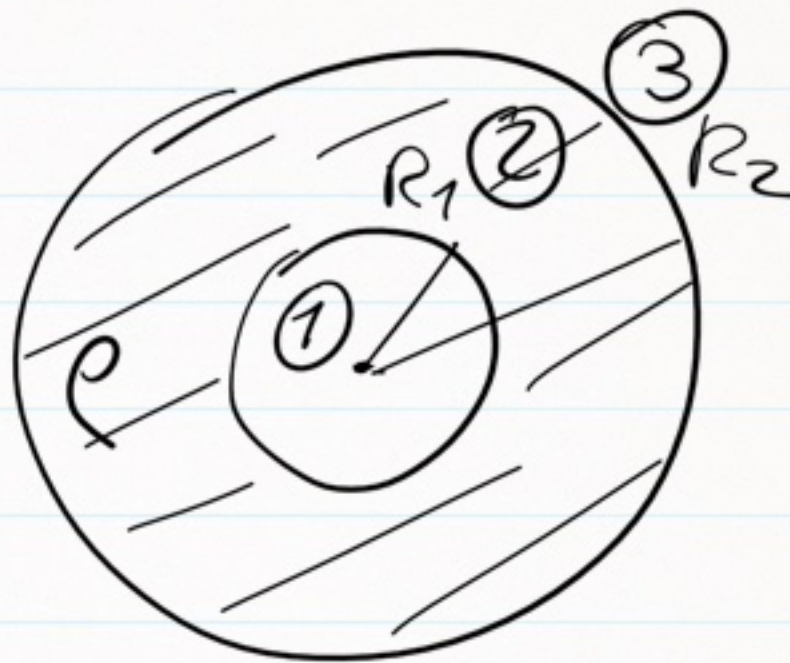
$$= \frac{\sigma}{\epsilon_0} \cdot \frac{R^2}{r}$$

$$r = R: \quad V(r=R) = \frac{\sigma R}{\epsilon_0}$$

$$r < R? \quad \vec{E} = -\vec{\nabla} V \quad ? \quad V(r) = \frac{\sigma R}{\epsilon_0}$$

$\downarrow$
 $\emptyset \rightarrow V = \text{cost.}$

③ Exercício 3



(a) $0 \leq r \leq R_1$: ① $E(r) = 0$

$$R_1 \leq r \leq R_2 : \textcircled{2} E(r) \cdot 4\pi r^2 = \int_{R_1}^r \frac{\rho \cdot 4\pi r'^2 dr'}{\epsilon_0}$$

$$= \frac{\rho}{\epsilon_0} 4\pi \left[\frac{r'^3}{3} \right]_{R_1}^r = \frac{\rho}{\epsilon_0} \frac{4\pi}{3} (r^3 - R_1^3)$$

$$\Rightarrow E(r) = \frac{\rho}{3\epsilon_0} \left(r - \frac{R_1^3}{r^2} \right)$$

$$r > R_2 : \textcircled{3} \quad E(r) \cdot 4\pi r^2 = \int_{R_1}^{R_2} \rho \cdot \frac{4\pi r'^2 dr'}{\epsilon_0}$$

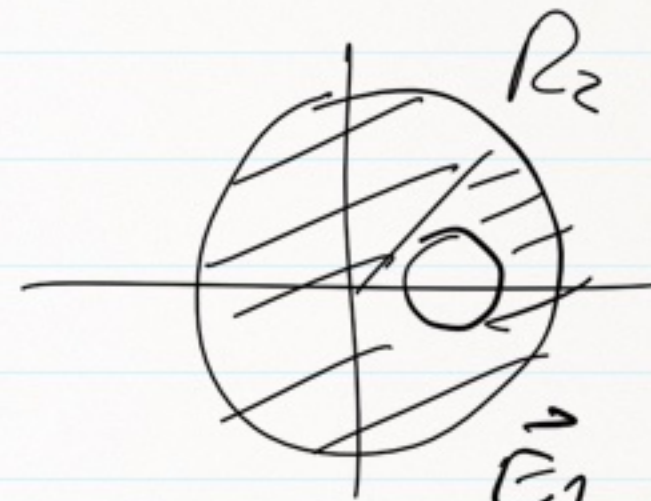
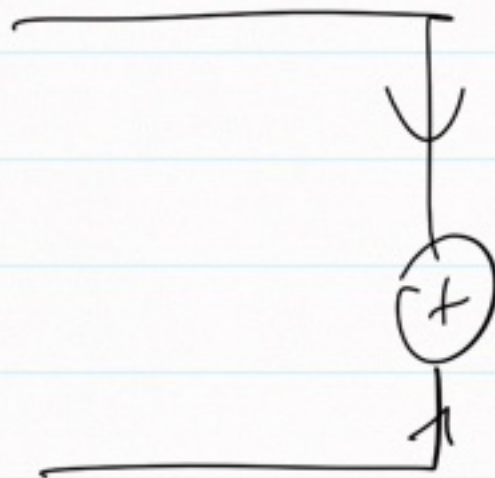
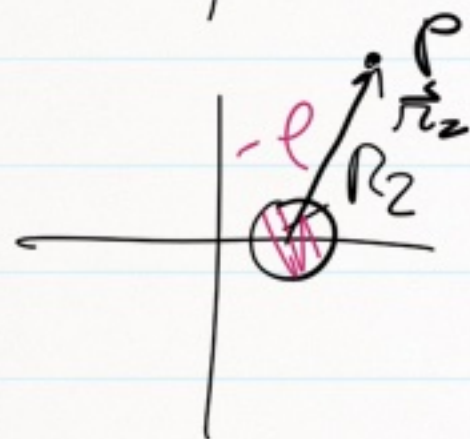
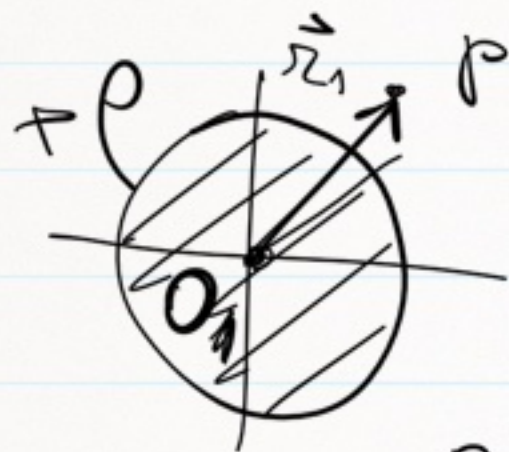
$$Q_3 = \rho \cdot \frac{4\pi}{3} (R_2^3 - R_1^3)$$

$$E(r) = \frac{\rho}{3\epsilon_0} \cdot \frac{R_2^3 - R_1^3}{r^2}$$

$$(b) \quad \text{ol ol p} \rightarrow V(R_1) - V(R_2) = ? \quad [100V]$$

$$\Delta V = \frac{\rho}{3\epsilon_0} \left[\frac{R_2^2 - R_1^2}{2} - R_1^3 \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \right] = 100V$$

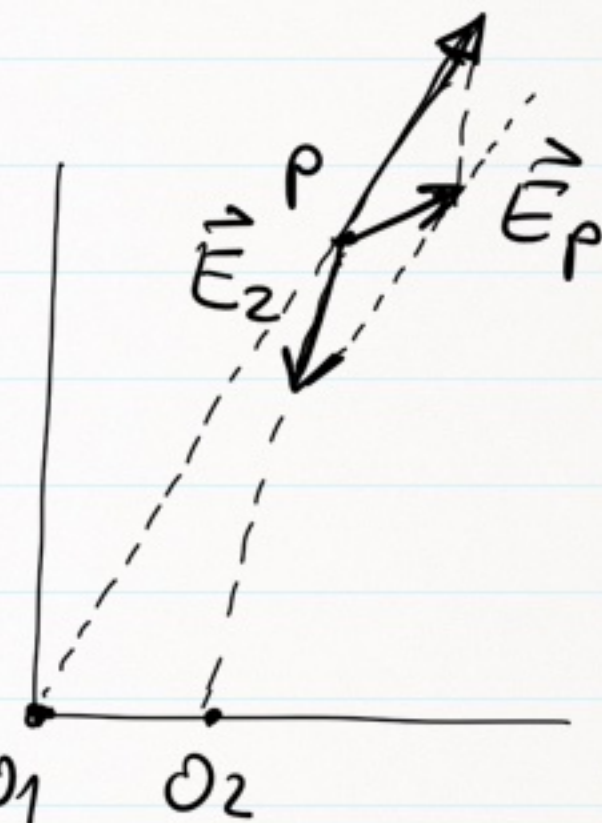
④ Esercizio 4



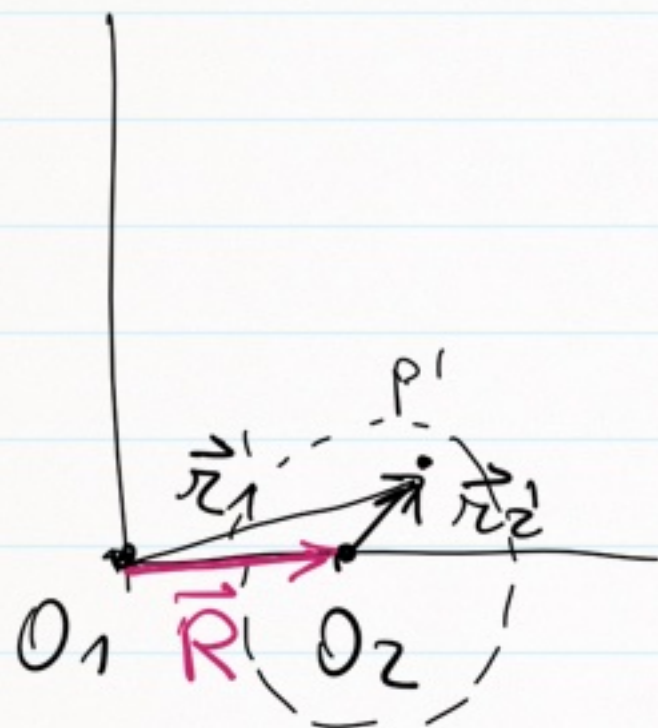
ρ : sphere 1 : $Q_1 = \rho \cdot \frac{4\pi}{3} R_1^3$

sphere 2 : $Q_2 = -\rho \cdot \frac{4\pi}{3} R_2^3$

$$\vec{E}_P = \vec{E}_1 + \vec{E}_2 = \frac{Q_1}{4\pi\epsilon_0 r_1^3} \cdot \vec{r}_1 + \frac{Q_2}{4\pi\epsilon_0 r_2^3} \cdot \vec{r}_2$$



|| ρ || :



$$q_1 = \rho \cdot \frac{4}{3} \pi r_1^3$$

$$q_2 = \rho \cdot \frac{4}{3} \pi r_2^3$$

$$\vec{E}_{P1} = \vec{E}_1 + \vec{E}_2 =$$

$$= \frac{1}{4 \pi \epsilon_0} \left(q_1 \frac{\vec{r}_1}{r_1^3} + q_2 \frac{\vec{r}_2}{r_2^3} \right)$$

$$= \frac{\rho}{3 \epsilon_0} (\vec{r}_1 - \vec{r}_2)$$

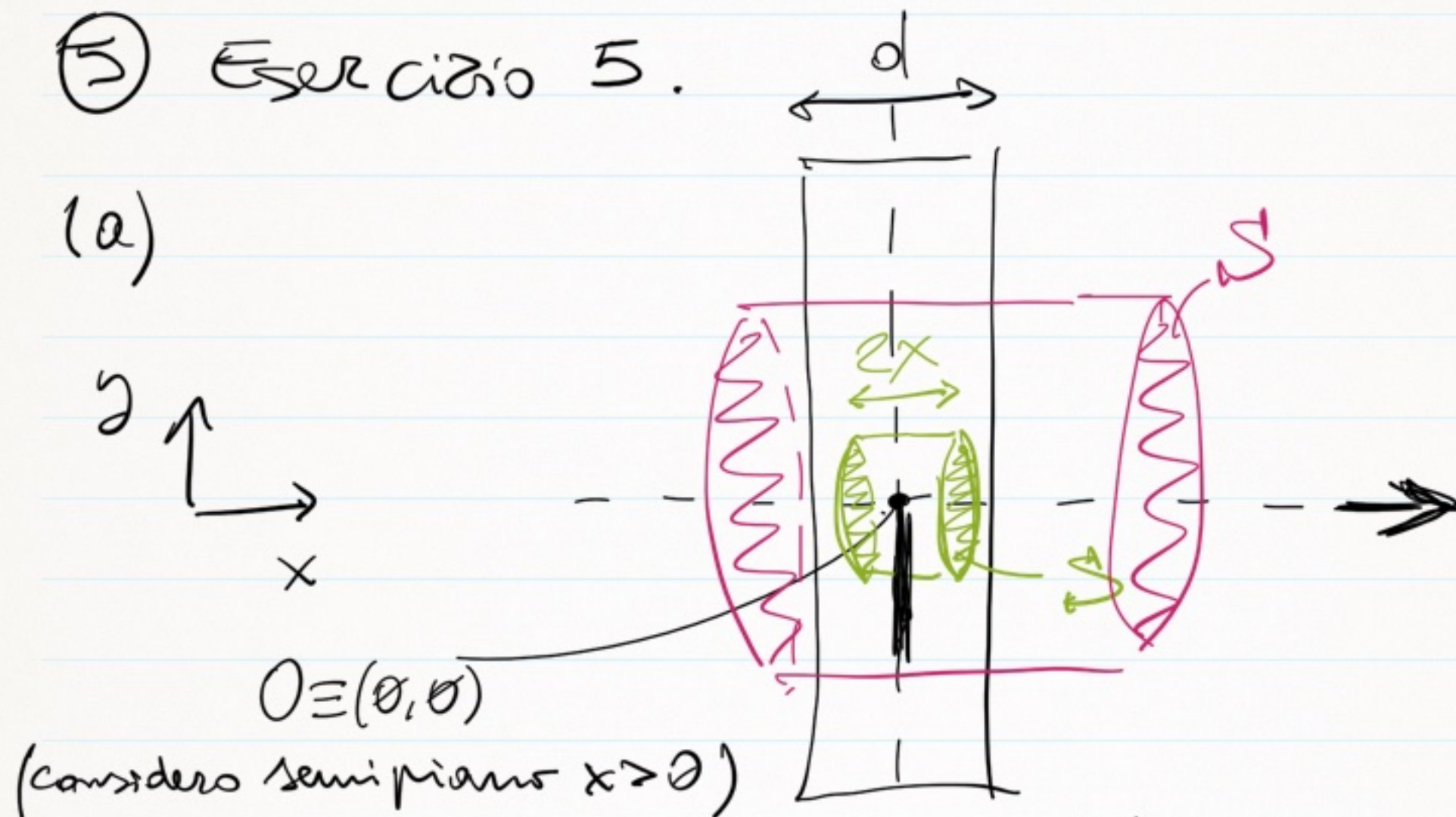
$$= \frac{\rho}{3 \epsilon_0} \vec{R}$$

(b) $\vec{F} = ?$ $\vec{E}_0$ vede la sfera forata come
SOVRAPPOSIZIONE di 2 cariche sferiche
EQUIVALENTI, in O_1 e O_2

$$\vec{F} = \frac{4}{3} \pi \rho (r_1^3 - r_2^3) \cdot \vec{E}_0$$

⑤ Esercizio 5.

1a)

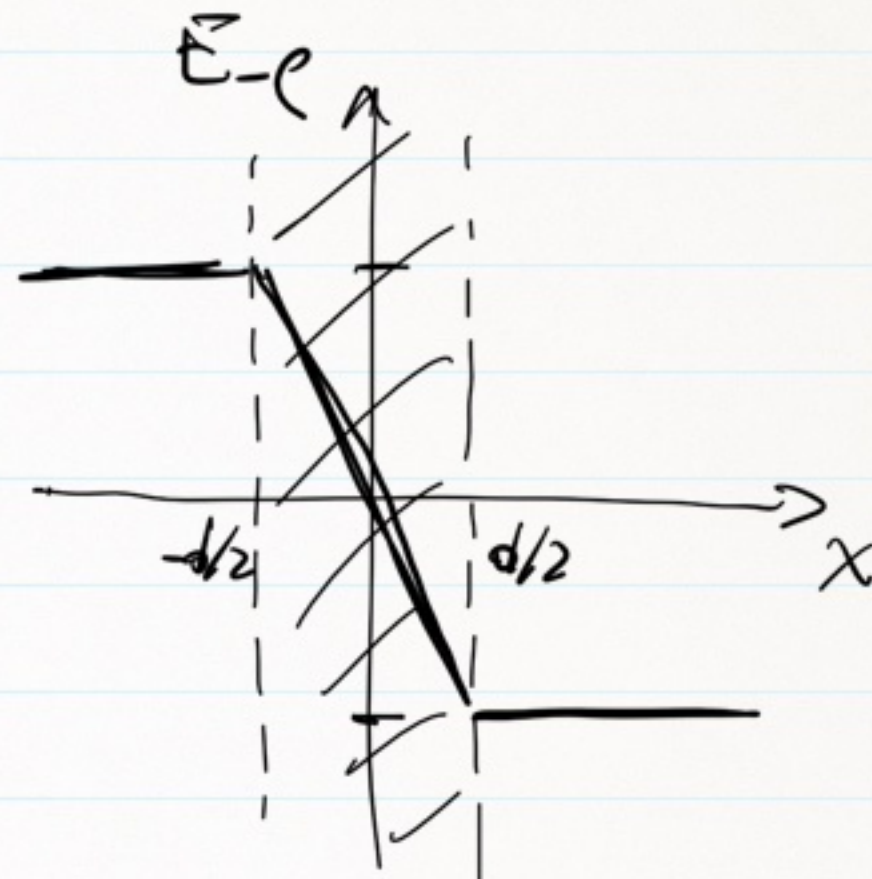
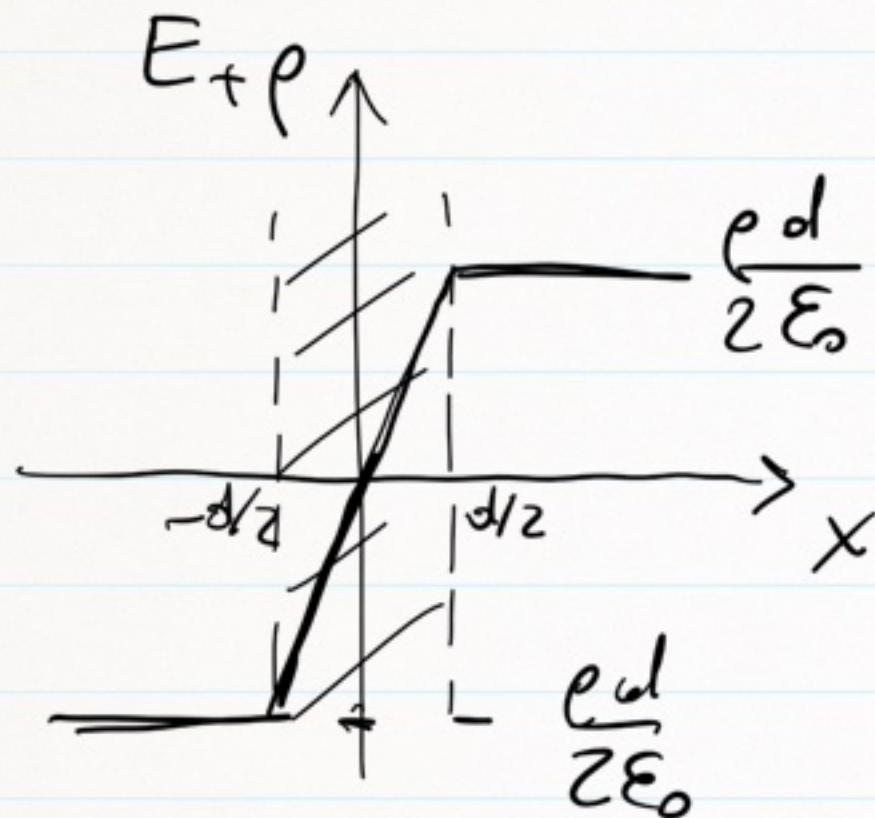


$O \equiv (0,0)$

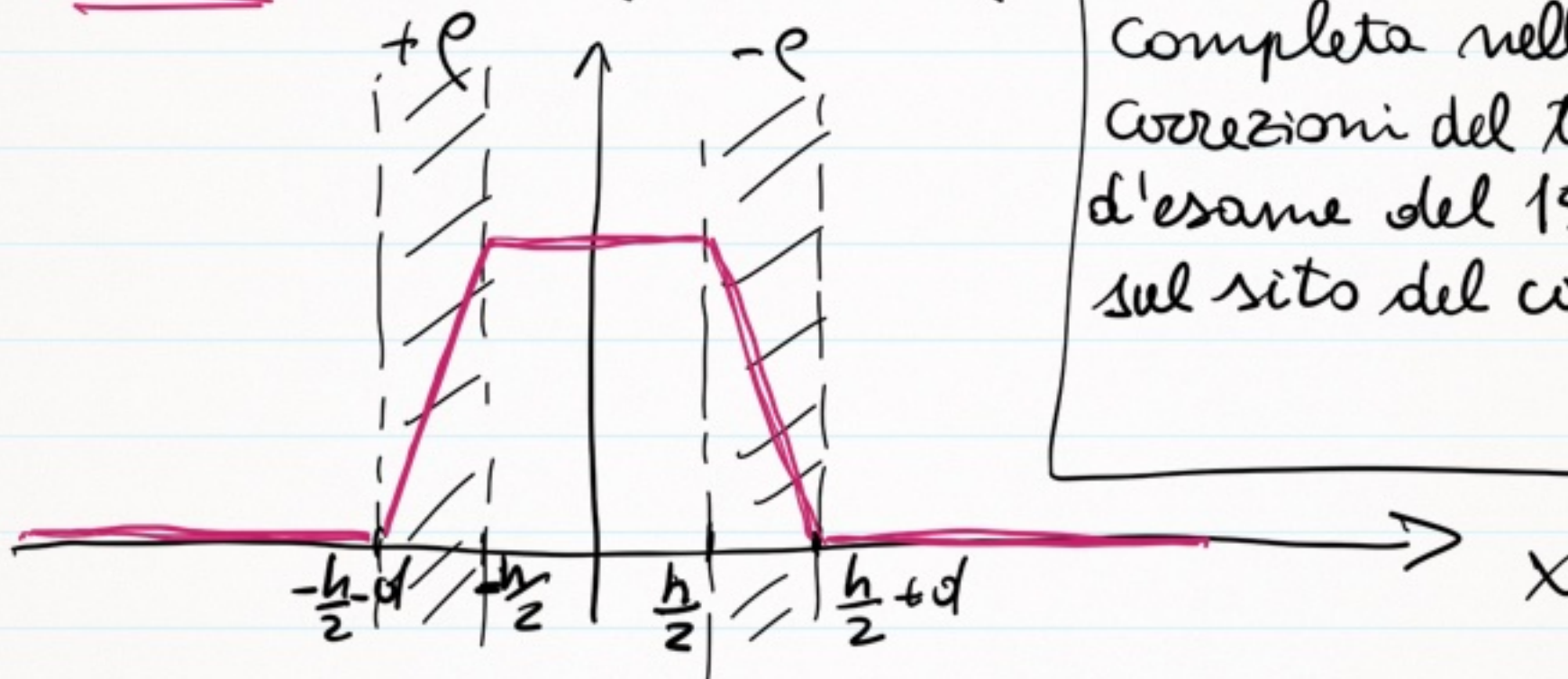
(considero semipiani $x > 0$)

$$\underline{x < d/2}: E(x) \cdot 2s = \frac{\rho \cdot 2x \cdot s}{\epsilon_0} \Rightarrow E(x) = \frac{\rho}{\epsilon_0} x$$

$$\underline{x > d/2}: E(x) \cdot 2s = \frac{\rho \cdot d \cdot s}{\epsilon_0} \Rightarrow E(x) = \frac{\rho d}{2\epsilon_0}$$



(b) $\underline{E_{tot}} = E + \rho + E - \rho$



Travate la soluzione completa nelle correzioni del temo d'esame del 15/06/20 sul sito del corso.